

QMST5336: Analytics

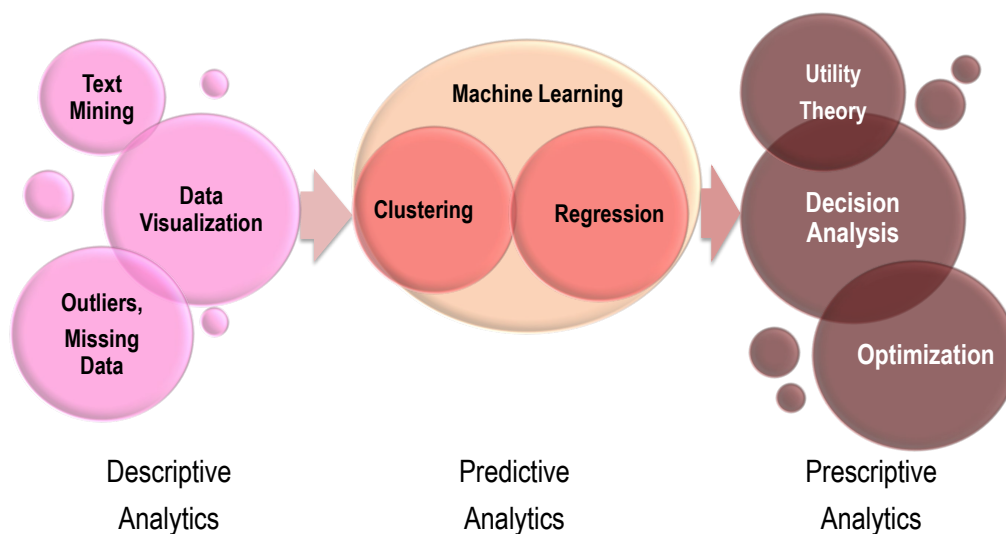
Lecture 2: Data Visualization

Emily Zhu, PhD.

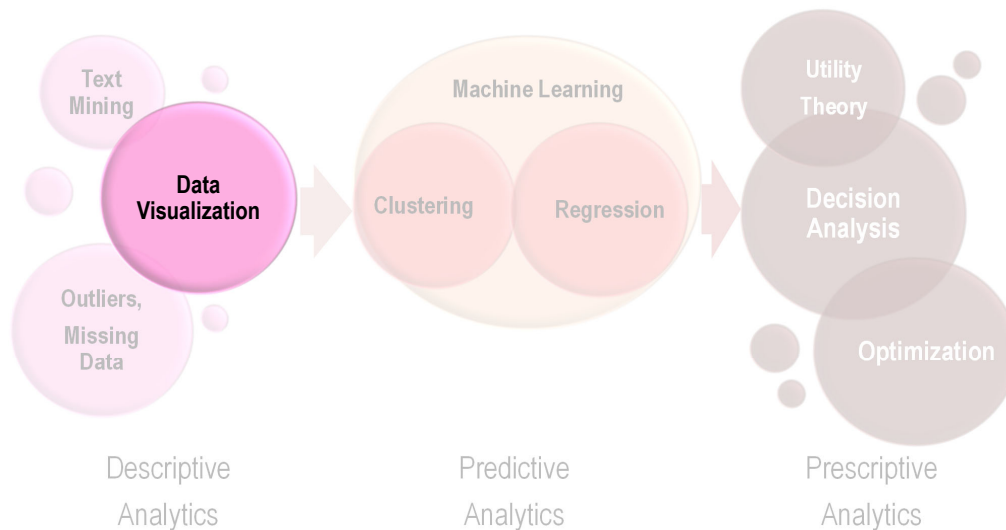
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Our Problem and Roadmap

An investment property in Austin: **Airbnb** versus **Rental**?



Roadmap: Where We Are



Why to Visualize the Data?

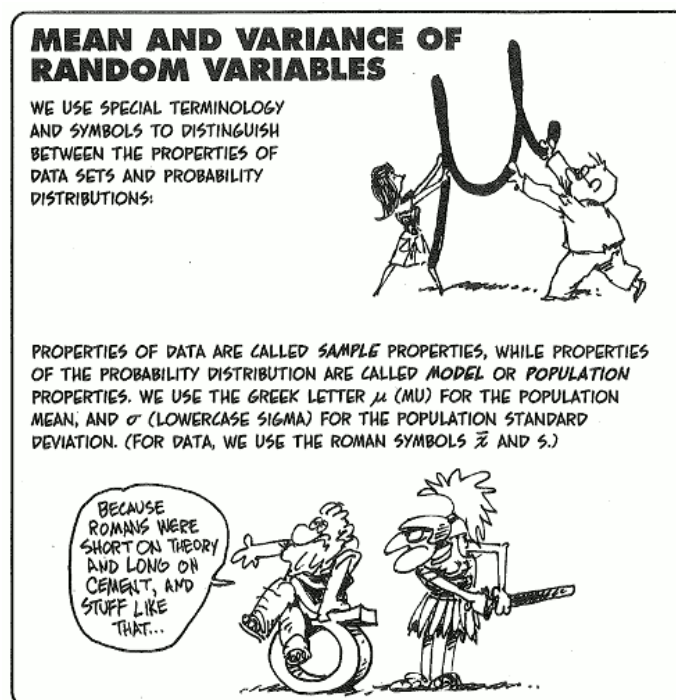
Reason 1: Get a full picture of the data



<https://www.theguardian.com/lifeandstyle/2008/nov/23/active-games-kids-guide-blindman-s-buff>

Why to Visualize the Data?

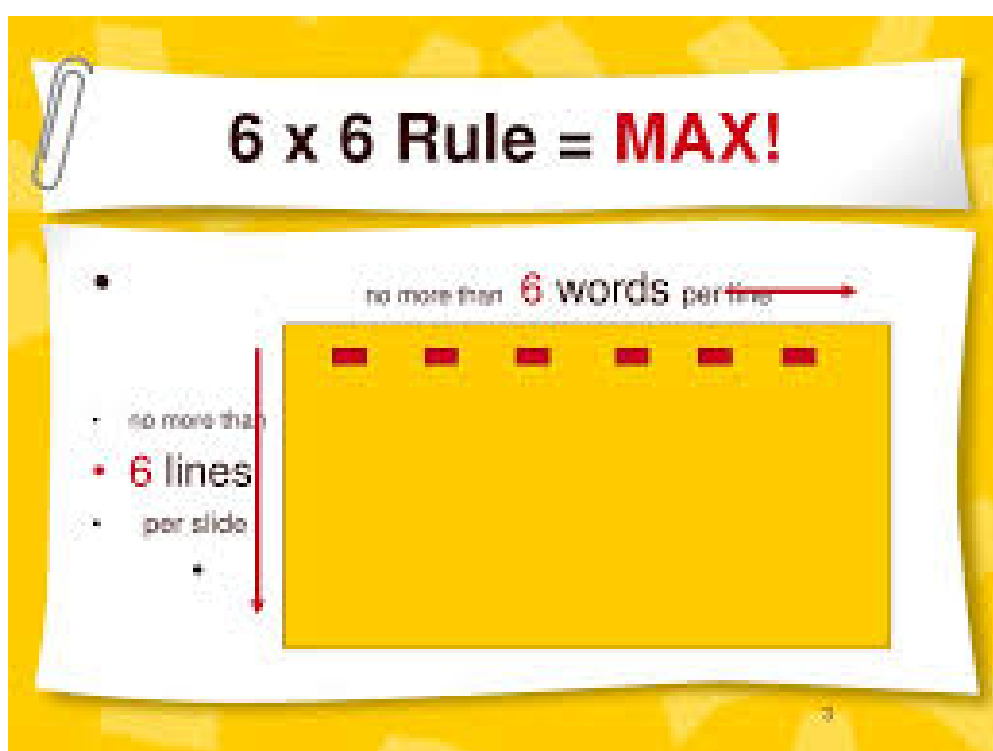
Reason 2: Find multiple key statistics in one shot



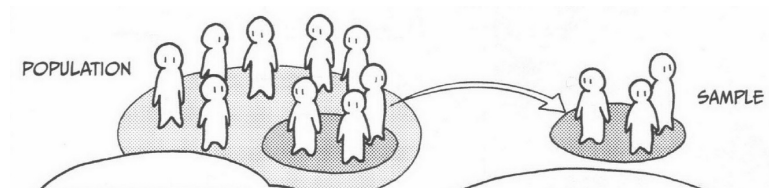
The cartoon guide to statistics

Why to Visualize the Data?

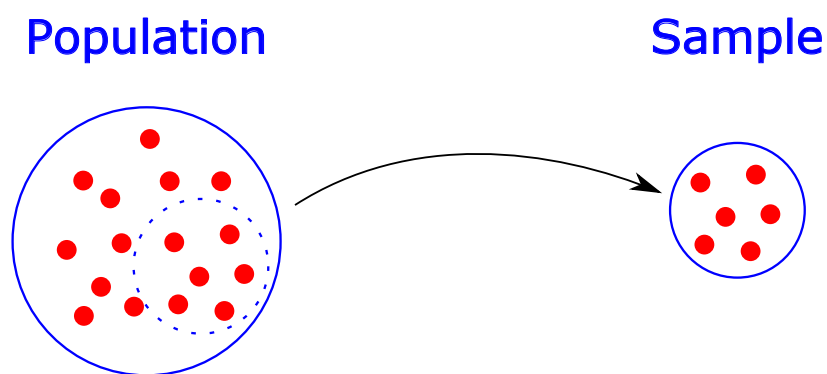
Reason 3: Make your presentation attractive



Basic Statistics



Population v.s. Sample



Example:

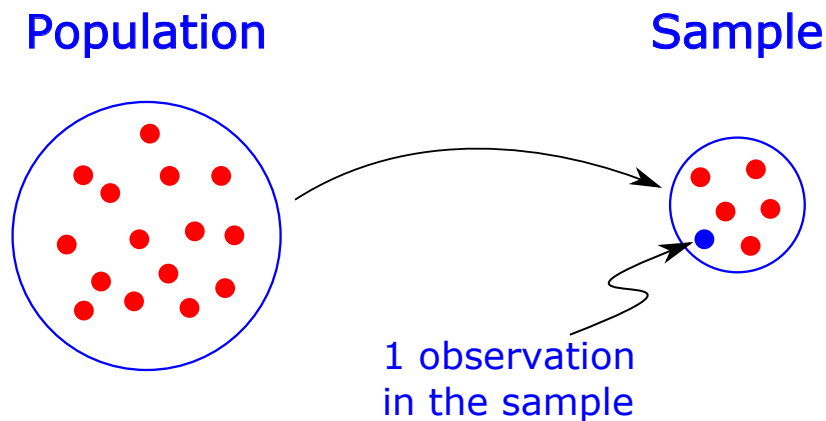
■ Population:

- All houses in Austin
- All financial advisors

■ Sample:

- One sample of 100 houses (from Zillow.com)
- One sample of 4 financial advisors

Population v.s. Sample: Observation



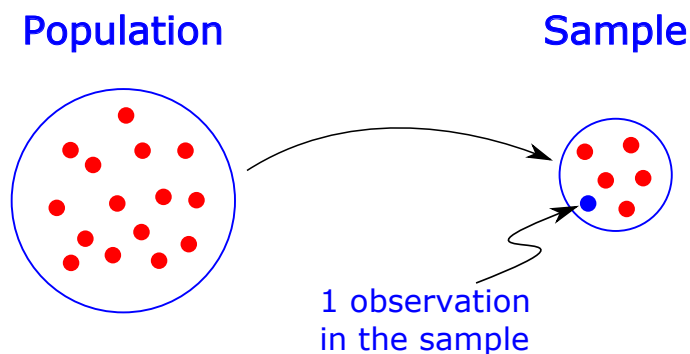
Observations:

- One observed house in Austin
- One observed financial advisor

Population v.s. Sample: Variables of an Observation

Variable: A characteristic of an observation

- There can be many variables describing an observation



Example: A house in Austin is 1 observation. Its Variables are:

- year built
- price
- number of bedrooms
- neighborhood
- type of property

Data Matrices

Postcode	Type	# bedroom	# bath	squareft	garage	land (arces)	HOA \$/month	year of construction	# Stroies	Price 000\$
78759	Condo	2	2	869	0	0.03	211	1982	2	213
78759	Condo	2	1	979	1	0.13	298	1985	1	289
78759	Condo	3	3	1616	2	0.07	0	1979	2	269
78759	House	3	2	1380	2	0.22	0	1981	1	390
78759	House	4	2.5	1914	2	0.2	0	1981	2	499
78759	House	4	2.5	2531	2	0.3	0	1975	2	605
78759	House	3	2	1369	1	0.19	0	1978	1	420
78759	Condo	3	2.5	1678	2	0.07	175	1972	2	524
78759	House	3	2	1392	2	0.18	0	1977	1	384.9

Observations & Variables in Data Matrices

Postcode	Type	# bedroom	# bath	squareft	garage	land (arces)	HOA \$/month	year of construction	# Stroies	Price 000\$
78759	Condo	2	2	869	0	0.03	211	1982	2	213
78759	Condo	2	1	979	1	0.13	298	1985	1	289
78759	Condo	3	3	1616	2	0.07	0	1979	2	269
78759	House	3	2	1380	2	0.22	0	1981	1	390
78759	House	4	2.5	1914	2	0.2	0	1981	2	499
78759	House	4	2.5	2531	2	0.3	0	1975	2	605
78759	House	3	2	1369	1	0.19	0	1978	1	420
78759	Condo	3	2.5	1678	2	0.07	175	1972	2	524
78759	House	3	2	1392	2	0.18	0	1977	1	384.9

Population v.s. Sample: Types of Variables

Variable: A characteristic of an observation

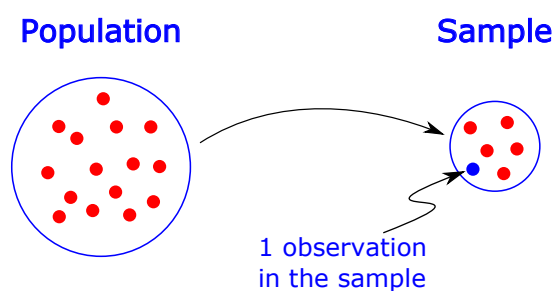
■ Quantitative Variable:

- Takes numerical values
- We can do arithmetic with them

■ Categorical Variable:

- Places an observation into categories
- We can't do arithmetic with them

Population v.s. Sample: Variables of an Observation



Example: An Airbnb in Austin is 1 observation. Its Variables are:

Quantitative Variables

- year of listing
- price
- number of guest

Categorical Variables

- neighborhood
- type of property
- type of room

Examining Quantitative Variables

Measures of Position:

- Refers to the central location of the data
- **Example:** what is the average price of a house in Austin?

Measures of Dispersion:

- Refers to how spread out (dispersed) the data is around the central location
- **Example:** do most houses cost the same, or do prices vary a lot?

Examining Quantitative Variables

Measures of Position:

- Mean
- Median
- Percentiles & quartiles

Measures of Dispersion:

- Variance
- Standard deviation
- Interquartile range

This is not an exhaustive list

Quantitative Data for Python

		monthly performance Experts (stock market)				monthly performance Monkeys (stock market)				Dow Jones Index	
	A	B	C	D	E	F	G	H	I	J	K
1	Month	Year	Expert #1	Expert #2	Expert #3	Expert #4	Monkey #1	Monkey #2	Monkey #3	Monkey #4	DJIA
2	Jul	1990	4.30	0.67	0.64	-1.39	4.48	-0.85	-0.72	-2.92	0.21
3	Aug	1990	4.73	3.15	2.32	-1.39	3.06	-0.31	-0.33	-1.83	0.96
4	Sep	1990	2.48	0.38	-0.78	-1.24	0.57	-0.82	-0.94	-3.58	-0.19
5	Oct	1990	-1.14	-1.52	-1.62	-2.38	3.70	-0.75	-1.69	-3.67	-0.77
6	Nov	1990	2.15	-3.32	-3.37	-8.08	1.08	-0.82	-2.62	-3.09	-0.71
7	Dec	1990	0.23	-1.44	-4.06	-5.83	-0.24	-2.01	-2.45	-4.45	-1.07
8	Jan	1991	0.74	0.05	-1.73	-2.44	-0.23	-1.92	-2.63	-2.73	-0.78
9	Feb	1991	0.42	-0.58	-1.18	-5.43	-0.23	-0.94	-3.33	-7.93	-0.07
10	Mar	1991	9.77	2.75	0.28	0.18	4.47	0.22	-0.13	-5.39	0.92
11	Apr	1991	6.18	3.98	0.41	-3.83	2.72	2.61	0.60	-2.19	1.32
12	May	1991	4.45	4.45	4.27	3.69	13.17	5.03	3.81	2.30	1.35
13	Jun	1991	7.24	6.93	4.88	3.28	2.36	2.28	1.23	1.17	1.44
14	Jul	1991	4.17	2.28	-1.00	-2.94	4.17	4.12	0.87	0.41	1.48
15	Aug	1991	4.17	1.33	0.41	-0.08	7.65	2.65	2.59	2.05	0.63
16	Sep	1991	16.11	1.06	-0.19	-3.78	22.69	1.80	-0.16	-0.58	0.37
17	Oct	1991	2.05	1.93	1.13	0.00	3.14	0.60	-0.28	-2.53	0.28
18	Nov	1991	4.43	1.00	-0.64	-0.66	14.60	0.50	-0.68	-0.76	0.37
19	Dec	1991	3.13	1.06	-1.26	-1.93	0.38	-1.00	-2.30	-4.83	-0.28
20	Jan	1992	1.98	1.16	1.11	-0.18	6.68	0.66	-2.03	-3.93	0.55

- Monthly data for ~ 12 years

Darts.csv

Examining Quantitative Variables

Measures of Position

A SMALL SET OF $n = 5$ DATA POINTS MAKES THE BOOKKEEPING EASY. SUPPOSE, FOR EXAMPLE, WE ASK FIVE PEOPLE HOW MANY HOURS OF TELEVISION THEY WATCH IN A WEEK... AND GET THE FOLLOWING ARRAY:

OBSERVATION	1	2	3	4	5
DATA VALUE	5	7	3	38	7

THEN $x_1 = 5$, $x_2 = 7$, $x_3 = 3$, $x_4 = 38$, AND $x_5 = 7$.

WHAT'S THE "CENTER" OF THESE DATA? THERE ARE ACTUALLY SEVERAL DIFFERENT WAYS TO MEASURE IT.



Measures of Position: Mean

Mean: It is the “center of mass”. Sometimes referred as the “average”

Sample Mean:

- You have a sample with n observations
- Observation 1 takes value x_1 , observation 2 takes value $x_2, \dots$
- Denote by $\bar{x}$ the **Sample Mean** given by

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Population Mean:

- You have a population with N observations
- Denote by μ the **Population Mean** given by

$$\mu = \frac{x_1 + x_2 + \dots + x_N}{N}$$

Measures of Position: Mean

Example: Suppose we have a sample of 8 houses

House	Price (thousands of dollars)
A	\$150
B	\$200
C	\$150
D	\$400
E	\$200
F	\$120
G	\$180
H	\$90
mean	\$186.25

$$\bar{x} = \frac{\$150 + \$200 + \$150 + \$400 + \$200 + \$120 + \$180 + \$90}{8} = \$186.25$$

Measures of Position: Mean

```
1 import numpy as np
2 L2 = df_full[['Expert #1', 'Monkey #1']]
3 x = np.mean(L2)
4 print(x)
```

Expert #1 4.078163

Monkey #1 3.485034

dtype: float64

In Excel: *AVERAGE()* gives you the mean

C152		=AVERAGE(C2:C148)										
	A	B	C	D	E	F	G	H	I	J	K	
143	Apr	2002	7.03	4.38	1.46	-0.18	0.96	0.95	-3.45	-4.40	1.06	
144	May	2002	1.40	1.22	-0.91	-1.05	4.24	-0.11	-1.18	-1.94	0.04	
145	Jun	2002	2.73	2.67	-0.23	-3.32	1.42	-0.23	-2.21	-3.00	0.10	
146	Jul	2002	2.98	-0.87	-0.99	-1.78	2.95	0.53	-5.69	-6.47	-0.58	
147	Aug	2002	1.19	-0.77	-1.75	-2.08	0.68	0.33	-1.25	-4.28	-1.01	
148	Sep	2002	-1.37	-4.21	-4.27	-5.09	1.95	-2.34	-2.84	-6.30	-1.47	
149												
150			Expert #1	Expert #2	Expert #3	Expert #4	Monkey #1	Monkey #2	Monkey #3	Monkey #4	DJIA	
151			Measures of location									
152			Mean	4.1	1.4	-0.2	-2.0	3.5	0.8	-0.7	-2.6	0.4

Lecture 02: Darts_Complete.xls

Measures of Position: Median

Median: Value at which

- 50% of the data lie above and
- 50% of the data lie below

Same definition for both Population and for Sample

To find median

- **Step 1:** sort observations by size
- **Step 2:**
 - If n is odd: median is middle observation
 - If n is even: median is the mean of two middle observations

Properties: Unlike the mean, the median is not affected by “outliers” (a few extreme values).

Measures of Position: Median

Example: a sample of 9 houses (*n* is odd)

House	Price
A	\$150
B	\$200
C	\$150
D	\$400
E	\$200
F	\$120
G	\$180
H	\$90
I	\$300

→ Sort →

House	Price
H	\$90
F	\$120
A	\$150
C	\$150
G	\$180
B	\$200
E	\$200
I	\$300
D	\$400

Median = \$180

Measures of Position: Median

Example: a sample of 8 houses (*n* is even)

House	Price
A	\$150
B	\$200
C	\$150
D	\$400
E	\$200
F	\$120
G	\$180
H	\$90

→ Sort →

House	Price
H	\$90
F	\$120
A	\$150
C	\$150
G	\$180
B	\$200
E	\$200
D	\$400

$$\text{Median} = \frac{\$150 + \$180}{2} = \$165$$

Measures of Position: Median

Example: Mean vs Median

Sample 1

House	Price
A	\$150
B	\$200
C	\$150
D	\$400
E	\$200
F	\$120
G	\$180
H	\$90

Sample 2

House	Price
A	\$150
B	\$200
C	\$150
D	\$900
E	\$200
F	\$120
G	\$180
H	\$90

Difference between Mean and Median

- For **Sample 1** the **mean** $\bar{x} = \$186.25$, and **median** \$165
- **Sample 2** is the same as **Sample 1** except for house D which is \$900
- For **Sample 2** the **mean** $\bar{x} = \$248.75$, and **median** \$165
- **Properties:** **Unlike the mean**, the **median** is not affected by outliers (a few extreme values).

Measures of Position: Median

```
1 import numpy as np
2 L2 = df_full[['Expert #1', 'Monkey #1']]
3 np.median(L2, axis = 0)
```

array([3.23, 2.29])

In Excel: *MEDIAN()* gives you the median

C153		=MEDIAN(C2:C148)									
147	A	B	C	D	E	F	G	H	I	J	K
Aug		2002	1.19	-0.77	-1.75	-2.08	0.68	0.33	-1.25	-4.28	-1.01
148	Sep		2002	-1.37	-4.21	-4.27	-5.09	1.95	-2.34	-2.84	-6.30
149											
150			Expert #1	Expert #2	Expert #3	Expert #4	Monkey #1	Monkey #2	Monkey #3	Monkey #4	DJIA
151			Measures of location								
152			Mean	4.1	1.4	-0.2	-2.0	3.5	0.8	-0.7	-2.6
153			Median	3.2	1.2	-0.2	-1.8	2.3	0.7	-0.4	-2.1
154			Percentile								

Measures of Position: Percentile

Percentiles: P_{th} percentile is the value below which $P\%$ of the observations lie

- 80_{th} Percentile → 80% of the observations lie below the value of P_{80}
- 20_{th} Percentile → 20% of the observations lie below the value of P_{20}

Same definition for both Population and for Sample

Measures of Position: Quartiles

Quartiles: There are three “special” percentiles. Sorting the data from the smallest to the largest value:

- Quartile 3: The 75th percentile
 - The “median” of the data above the overall median
- Quartile 2: The 50th percentile (the median)
- Quartile 1: The 25th percentile
 - The “median” of the data below the overall median

Measures of Position: Quartiles

Example: a sample of $n = 8$ houses

House	Price
H	\$400
F	\$200
A	\$200
C	\$180
G	\$150
B	\$150
E	\$120
D	\$90

$$\text{Quartile 3} = \frac{\$200 + \$200}{2} = \$200$$

$$\text{Quartile 2} = \text{Median} = \frac{\$150 + \$180}{2} = \$165$$

$$\text{Quartile 1} = \frac{\$120 + \$150}{2} = \$135$$

Measures of Position: Percentile

```
1 L2 = df_full[['Expert #1', 'Monkey #1']]
2 np.percentile(L2, 25, axis = 0)
```

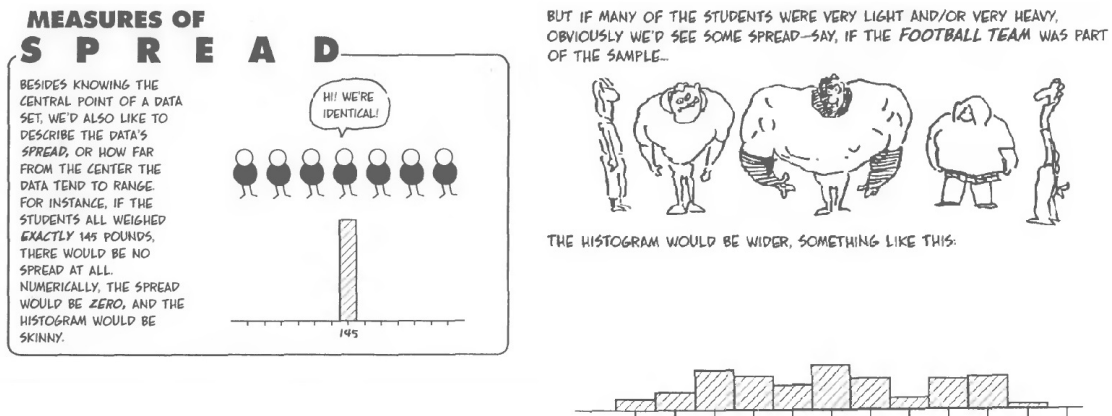
array([1.59 , 1.235])

In Excel: *PERCENTILE.INC*(array, $p\%$) & *PERCENTILE.EXC*(array, $p\%$) gives you the percentile P

C158 : =PERCENTILE(C2:C148,0.9)												
	A	B	C	D	E	F	G	H	I	J	K	
146	Jul	2002	2.98	-0.87	-0.99	-1.78	2.95	0.53	-5.69	-6.47	-0.58	
147	Aug	2002	1.19	-0.77	-1.75	-2.08	0.68	0.33	-1.25	-4.28	-1.01	
148	Sep	2002	-1.37	-4.21	-4.27	-5.09	1.95	-2.34	-2.84	-6.30	-1.47	
149												
150			Expert #1	Expert #2	Expert #3	Expert #4	Monkey #1	Monkey #2	Monkey #3	Monkey #4	DJIA	
151			Measures of location									
152			Mean	4.1	1.4	-0.2	-2.0	3.5	0.8	-0.7	-2.6	0.4
153			Median	3.2	1.2	-0.2	-1.8	2.3	0.7	-0.4	-2.1	0.4
154			Percentile									
155			10%	0.42	-0.84	-2.82	-4.71	0.54	-0.96	-2.62	-6.12	-0.54
156		(quartile 1) 25%	1.59	-0.04	-1.21	-3.59	1.24	-0.23	-1.66	-4.13	-0.03	
157		(quartile 3) 75%	5.44	2.64	0.91	-0.65	3.65	2.01	0.37	-1.00	0.97	
158		90%	8.534	3.886	1.988	0.232	6.916	2.708	1.15	-0.12	1.434	

Examining Quantitative Variables

Measures of Dispersion



Cartoon Guide to Statistics

Measures of Dispersion: Variance

Variance: Roughly speaking, it **measures** how far a set of observations are **spread** out **from** their **mean value**.

- If the observations are crowded: variance is small.
- If the observations are spread out: variance is large.

The **Sample Variance** for a sample with **n observations** is denoted by s^2 , where $\bar{x}$ is the sample mean:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

The **Population Variance** for a population with **N observations** is denoted by σ^2 , where μ is the population mean:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Measures of Dispersion: Standard Deviation

Standard Deviation: The square root of the variance

The **Sample Standard Deviation** is denoted by s

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

The **Population Standard Deviation** is denoted by σ

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

- Standard Deviation has the same units as the data
- Variance, intuitively speaking, has units (units of data)²

Measures of Dispersion: Variance

Example: Sample with $n = 8$ observations

House	Price	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
A	\$150		
B	\$200		
C	\$150		
D	\$400		
E	\$200		
F	\$120		
G	\$180		
H	\$90		

$$\bar{x} = \frac{1}{8} \sum_{i=1}^8 x_i = \$186.25,$$

Measures of Dispersion: Variance

Example: Sample with $n = 8$ observations

House	Price	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
A	\$150	\$-36.25	
B	\$200	\$13.75	
C	\$150	\$-36.25	
D	\$400	\$213.75	
E	\$200	\$13.75	
F	\$120	\$-66.25	
G	\$180	\$-6.25	
H	\$90	\$-96.25	

$$\bar{x} = \frac{1}{8} \sum_{i=1}^8 x_i = \$186.25,$$

Measures of Dispersion: Variance

Example: Sample with $n = 8$ observations

House	Price	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
A	\$150	\$-36.25	1314.0625
B	\$200	\$13.75	189.0625
C	\$150	\$-36.25	1314.0625
D	\$400	\$213.75	45689.0625
E	\$200	\$13.75	189.0625
F	\$120	\$-66.25	4389.0625
G	\$180	\$-6.25	39.0625
H	\$90	\$-96.25	9264.0625

$$\bar{x} = \frac{1}{8} \sum_{i=1}^8 x_i = \$186.25, \quad \sum_{i=1}^8 (x_i - \bar{x})^2 = 62387.5,$$

Measures of Dispersion: Variance

Example: Sample with $n = 8$ observations

House	Price	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
A	\$150	\$-36.25	1314.0625
B	\$200	\$13.75	189.0625
C	\$150	\$-36.25	1314.0625
D	\$400	\$213.75	45689.0625
E	\$200	\$13.75	189.0625
F	\$120	\$-66.25	4389.0625
G	\$180	\$-6.25	39.0625
H	\$90	\$-96.25	9264.0625

$$\bar{x} = \frac{1}{8} \sum_{i=1}^8 x_i = \$186.25, \quad \sum_{i=1}^8 (x_i - \bar{x})^2 = 62387.5,$$

$$s^2 = \frac{1}{7} \sum_{i=1}^8 (x_i - \bar{x})^2 = 8912.5, \quad s = \sqrt{\frac{1}{7} \sum_{i=1}^8 (x_i - \bar{x})^2} \approx \$94.4$$

Measures of Dispersion: Variance

```

1 import numpy as np
2 L2 = df_full[['Expert #1', 'Monkey #1']]
3 np.std(L2, ddof = 1) #sample standard deviation
4 np.std(L2) #population

```

Expert #1 3.836682

Monkey #1 5.176784

dtype: float64

In Excel:

- **VAR.S()** gives you the sample variance.
- **STDEV.S()** gives you the sample standard deviation.

C163		:	<i>f_x</i>		=VAR.S(C2:C148)							
	A	B	C	D	E	F	G	H	I	J	K	
146	Jul	2002	2.98	-0.87	-0.99	-1.78	2.95	0.53	-5.69	-6.47	-0.58	
147	Aug	2002	1.19	-0.77	-1.75	-2.08	0.68	0.33	-1.25	-4.28	-1.01	
148	Sep	2002	-1.37	-4.21	-4.27	-5.09	1.95	-2.34	-2.84	-6.30	-1.47	
149												
150			Expert #1	Expert #2	Expert #3	Expert #4	Monkey #1	Monkey #2	Monkey #3	Monkey #4	DJIA	
151			Measures of location									
152			Mean	4.1	1.4	-0.2	-2.0	3.5	0.8	-0.7	-2.6	0.4
153			Median	3.2	1.2	-0.2	-1.8	2.3	0.7	-0.4	-2.1	0.4
154			Percentile									
155			10%	0.42	-0.84	-2.82	-4.71	0.54	-0.96	-2.62	-6.12	-0.54
156		(quartile 1) 25%	1.59	-0.04	-1.21	-3.59	1.24	-0.23	-1.66	-4.13	-0.03	
157		(quartile 3) 75%	5.44	2.64	0.91	-0.65	3.65	2.01	0.37	-1.00	0.97	
158		90%	8.534	3.886	1.988	0.232	6.916	2.708	1.15	-0.12	1.434	

Measures of Dispersion: Interquartile Range

Interquartile Range (IQR): The range of the middle 50% of the observations in a data set.

- 75th percentile minus the 25th percentile, or
- 3rd quartile minus 1st quartile (Quartile 3 - Quartile 1)

Same definition for both Population and for Sample

Properties: Unlike the variance or the standard deviation, the percentiles (and hence the Interquartile Range) are not affected by outliers.

Measures of Dispersion: Interquartile Range

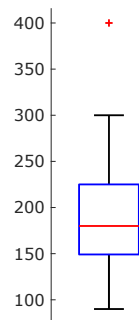
```
1 from scipy import stats
2 L2 = df_full[['Expert #1', 'Monkey #1']]
3 stats.iqr(L2, axis = 0)

array([3.85 , 2.415])
```

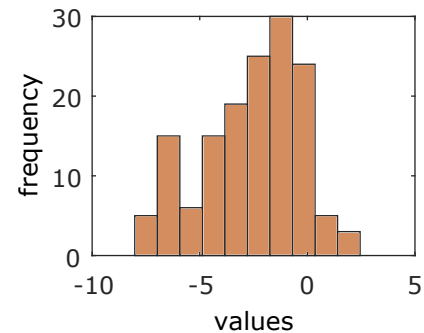
Visualize One Numerical Variable

Graphical Analysis

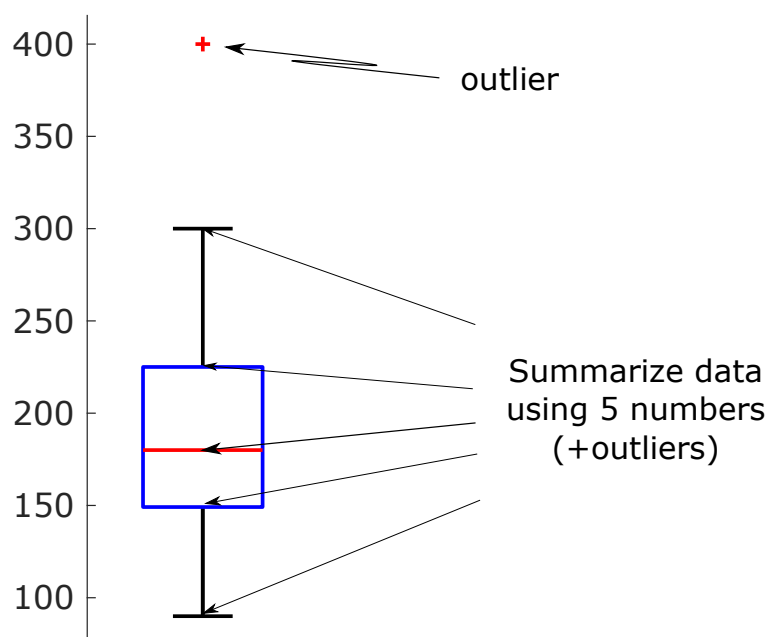
Box Plot



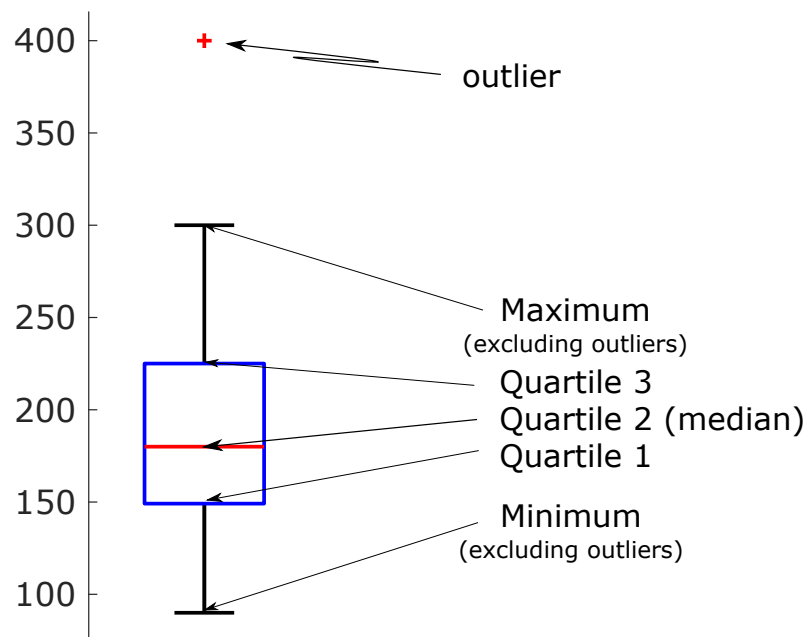
Histogram



Graphical Analysis: Box Plots



Graphical Analysis: Box Plots



Graphical Analysis: Box Plots

Box Plot: Represents graphically the general shape of a distribution. The graph contains

- Minimum value of the observation (excluding outliers)
- Quartile 1
- Quartile 2 (median)
- Quartile 3
- Maximum value of the observation (excluding outliers)
- Outliers (sometimes if necessary)

An observation is considered an outlier if

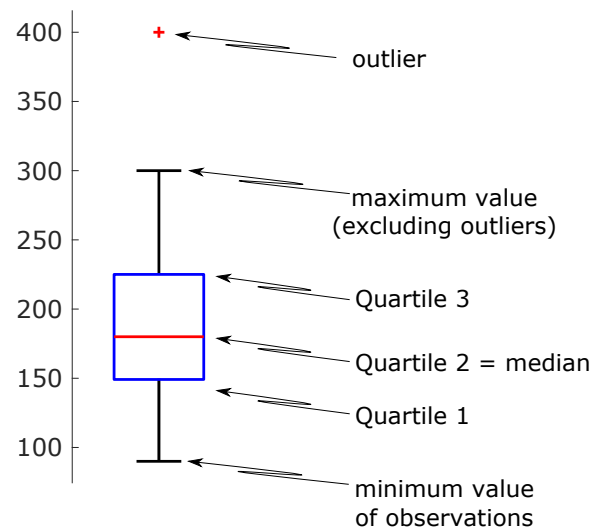
- it has value $\geq Q3 + 1.5(Q3 - Q1) = Q3 + 1.5(IQR)$, or
- it has value $\leq Q1 - 1.5(Q3 - Q1) = Q1 - 1.5(IQR)$

Some textbooks use a different value than 1.5

Graphical Analysis: Box Plots

Example: a sample of $n = 9$ houses

Samples		Sorted Samples	
House	Price	House	Price
A	\$150	D	\$400
B	\$200	I	\$300
C	\$150	E	\$225
D	\$400	B	\$200
E	\$225	G	\$180
F	\$120	A	\$150
G	\$180	C	\$150
H	\$90	F	\$120
I	\$300	H	\$90



Quartile 1 = \$150, Quartile 2 = \$180, Quartile 3 = \$225, IQR = \$75

$$\$400 \geq Q3 + 1.5(IQR) = \$225 + 1.5 \times \$75 = \$337.5$$

Graphical Analysis: Box Plots

```
1 import matplotlib.pyplot as plt
2 plt.boxplot(df_full['Expert #1'])
3 plt.show()
```

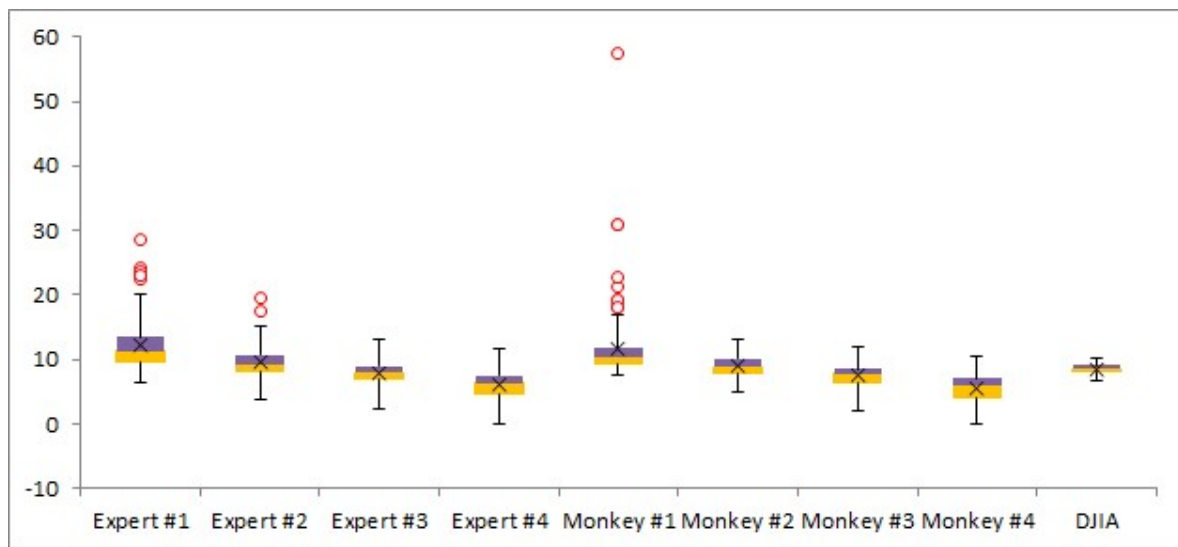
In Excel: You can plot box plots in Excel **insert** and then click **Box and Whisker**.

To generate the boxplots for multiple groups,

- press **Ctrl-m** and select the **Descriptive Statistics and Normality** data analysis tool.
- in the dialog box that appears, select the **Box Plot** option and insert data in the **Input Range**.
- check **Headings included with the data** and uncheck **Use exclusive version of quartile**.

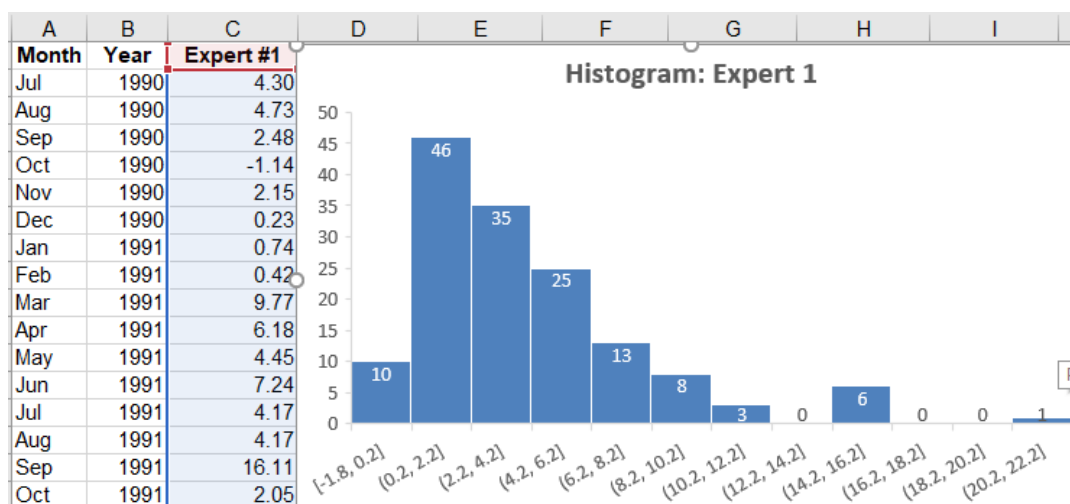
Graphical Analysis: Box Plot

Example: Investors and Monkeys



Graphical Analysis: Histogram

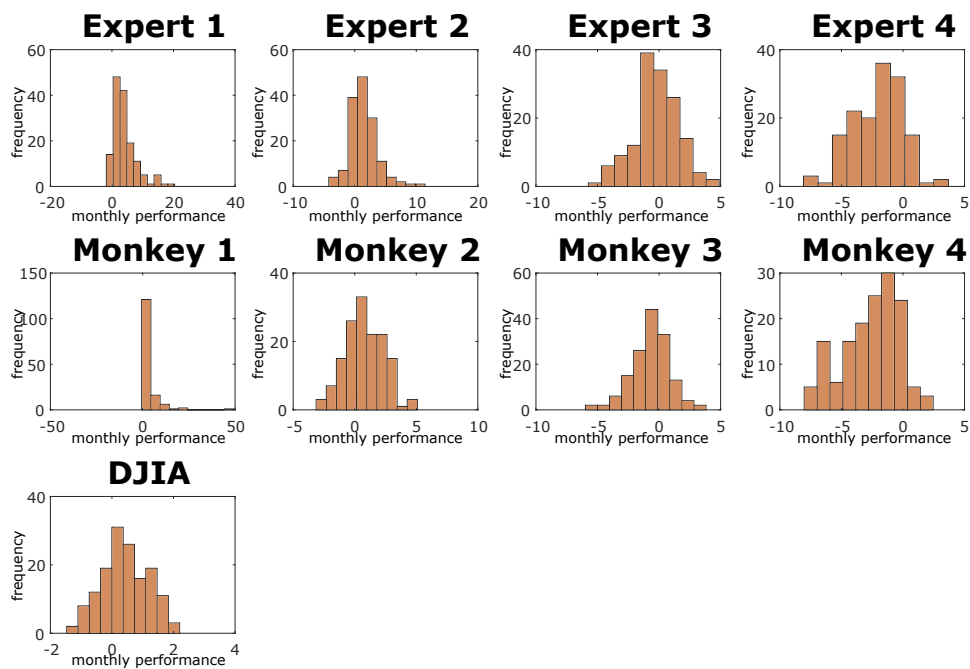
Example: Investors and Monkeys



Graphical Analysis: Histogram

Histogram:

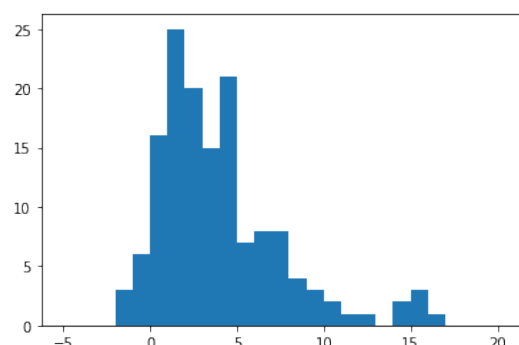
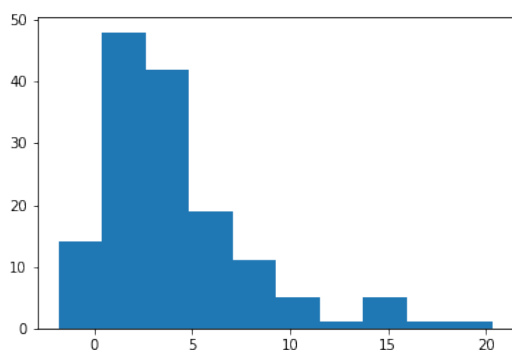
- Value of the variable is plotted on the horizontal axis
- Frequency is plotted on the vertical axis



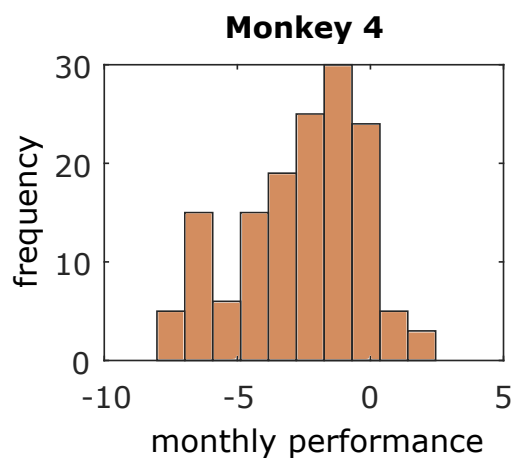
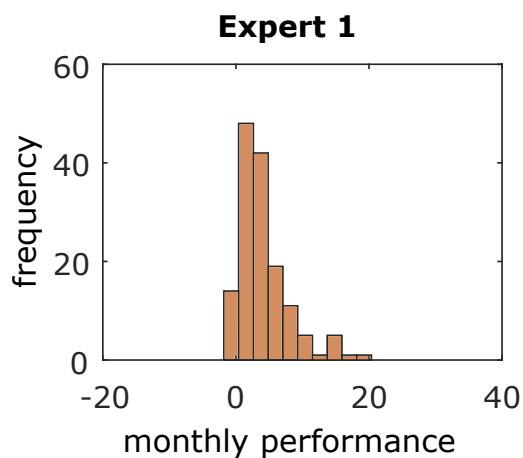
Graphical Analysis: Histogram

```
1 import matplotlib.pyplot as plt
2 plt.hist(df_full['Expert #1'])
3 plt.show()
4 #Another way
5 df_full['Expert #1'].hist()
```

```
1 #customize bin widths
2 bin_width = 1
3 plt.hist(df_full['Expert #1'],
4         bins = range(-5, 21,
5                     bin_width))
6 #range(min, max, bin_width)
7 #Another way
8 df_full['Expert #1'].hist(bins
9                          = range(-5, 21, bin_width
10                          ))
```



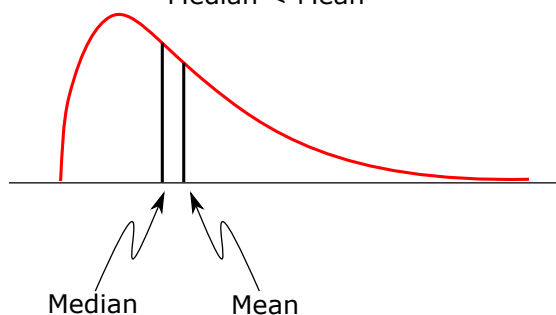
Skewness



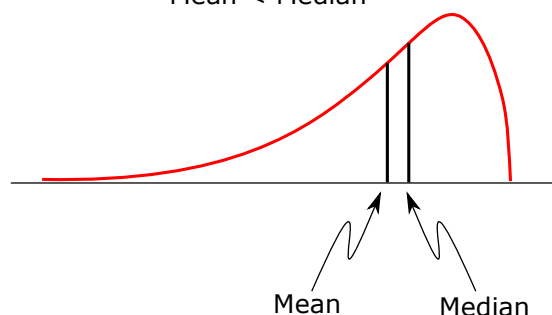
Skewness

Skewness: Refers to the shape of the distribution of values

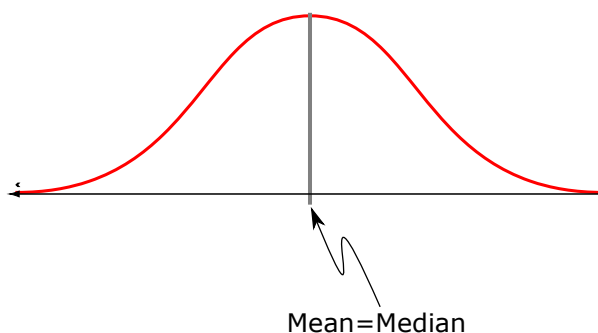
Positively Skewed (Skewed to the **Right**)
 $\text{Median} < \text{Mean}$



Negatively Skewed (Skewed to the **Left**)
 $\text{Mean} < \text{Median}$



Distribution is **Symmetric**
 $\text{Mean} = \text{Median}$



Skewness

```
1 from scipy import stats
2 L2 = df_full[['Expert #1', 'Monkey #1']]
3 stats.skew(L2)
```

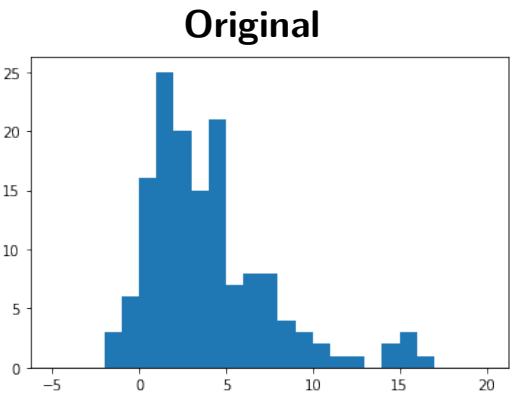
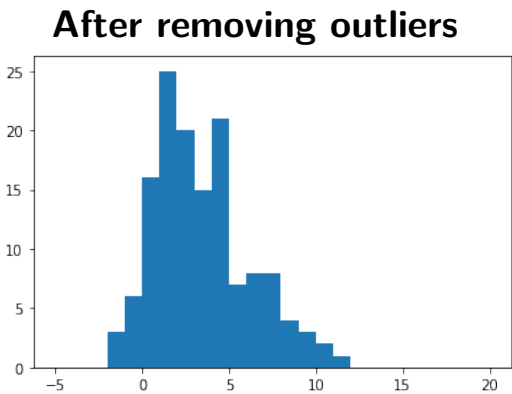
array([1.574426 , 5.64325181])
In Excel: *SKEW()* gives you the skewness.

- *SKEW()* is used for the Sample, and *SKEW.P()* for the Population

C168	:				=SKEW(C2:C148)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
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Impacts of Outliers

```
1 #visualize
2 bin_width = 1
3 plt.hist(df['Expert #1'], bins
4         = range(-5, 21, bin_width)
5         )
6 plt.show()
```



Examining Two Numerical Variables

What is the relationship between two variables?

House	Price	Bedrooms
A	\$150	1
B	\$200	3
C	\$150	2
D	\$400	5
E	\$200	2
F	\$120	1
G	\$180	4
H	\$90	1

Examining Relationship Visually: Scatter Plot

Scatter Plot:

- Let $x_1, \dots, x_n$ be the values of the first quantitative variable (explanatory variable)
- Let $y_1, \dots, y_n$ be the values of the second quantitative variable (response variable)
- The scatter plot displays the two quantitative variables as pairs of data: $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

Note: It is your job to decide

- which is the explanatory variable (x-axis)
- which is the response variable (y-axis)

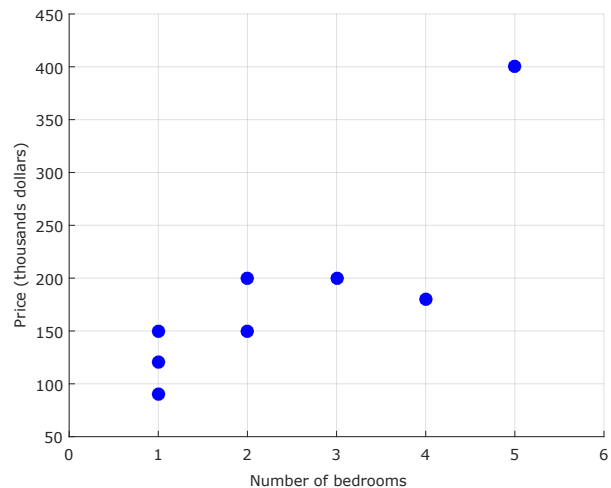
i.e., which is the *explanatory variables* that might affect *response variables*.

Examining Relationship Visually

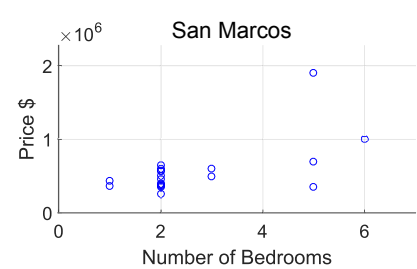
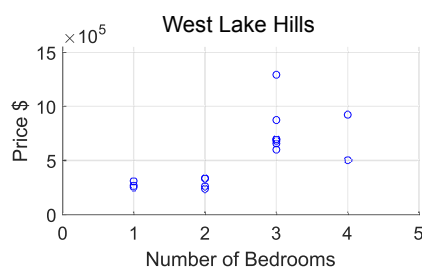
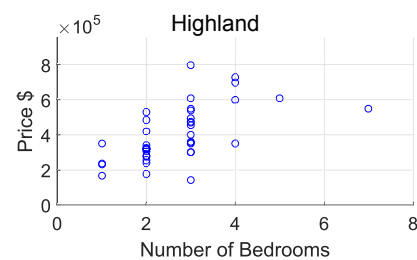
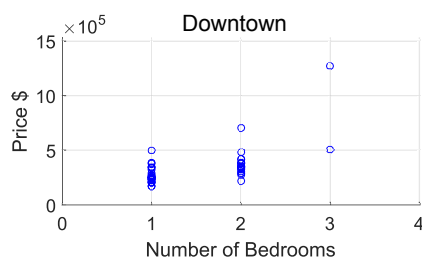
Example: Explain how the **price is affected by the number of bedrooms**

- explanatory variable “Number of bedrooms” (x-axis)
- Response variable “Price” (y-axis)

House	Price	Bedrooms
A	\$150	1
B	\$200	3
C	\$150	2
D	\$400	5
E	\$200	2
F	\$120	1
G	\$180	4
H	\$90	1



Examining Relationship Visually: Scatter Plot



Scatter Plot

```
1 #scatterplot
2 import matplotlib.pyplot as plt
3 plt.scatter(df_full['Bedrooms'], df_full['Price(000$)'])
4 plt.show()
```

What does a scatter plot present?

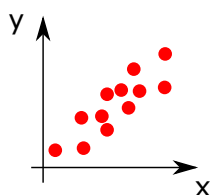
- Direction of association
- Strength of relationship
- Outliers
- Skewness

Scatter Plot: Direction of association

Direction of association:

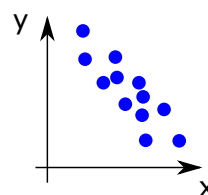
- The linear association is called Correlation

Positive Correlation



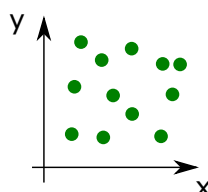
There is a **positive linear relationship** between x and y:
As x increases, y increases

Negative Correlation



There is a **negative linear relationship** between x and y:
As x increases, y decreases

No Correlation



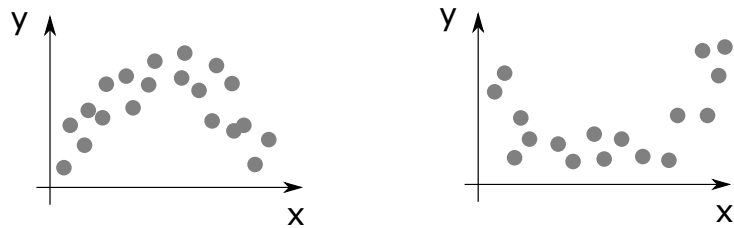
There is **no linear relationship** between x and y

Scatter Plot: Direction of association

Direction of association:

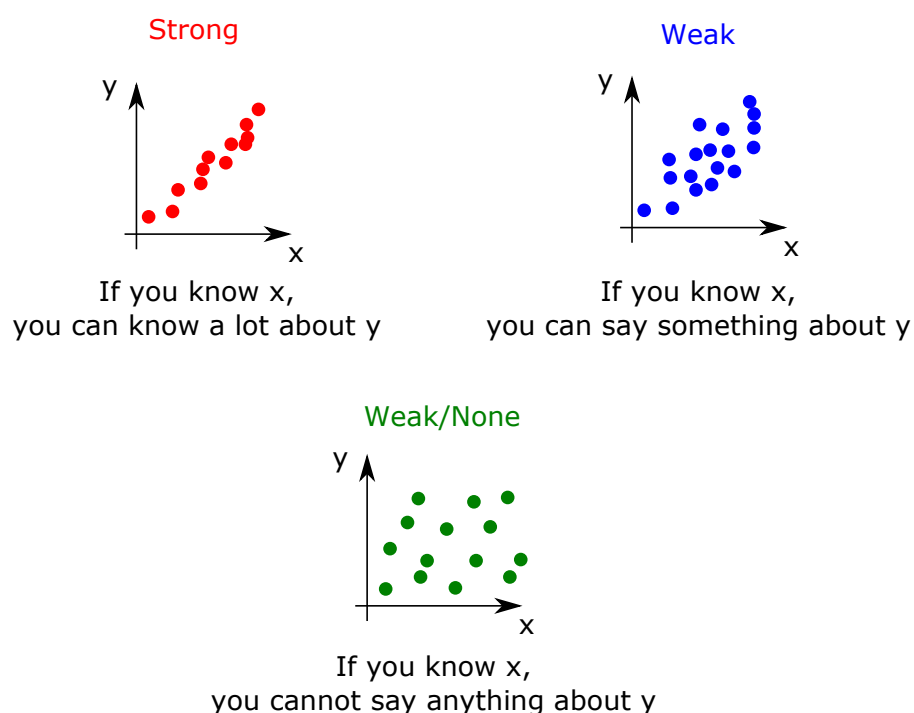
- There might be non-linear association between the variables

Non-Linear relationship



Scatter Plot: Strength of relationship

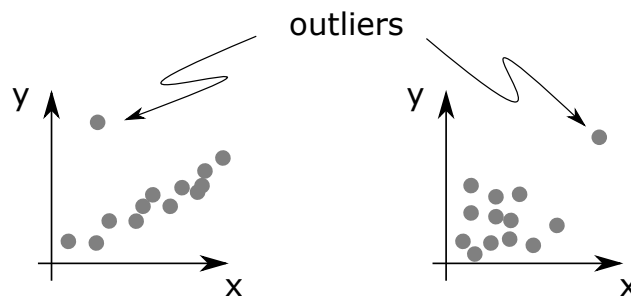
Strength of relationship:



Scatter Plot: Outliers

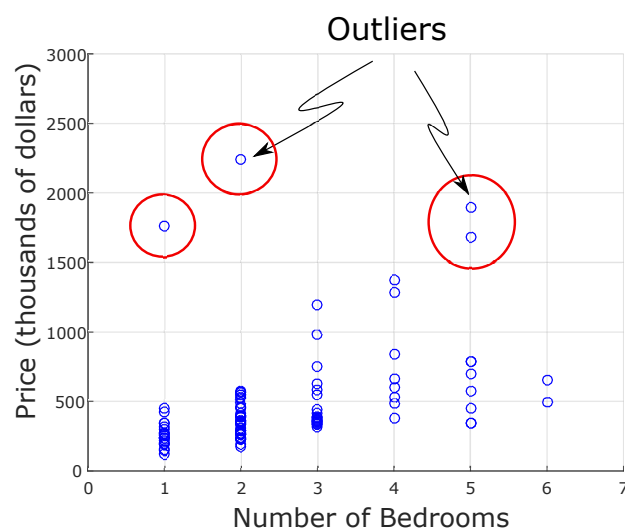
Outliers:

- An observation that has a low probability of occurrence
- Unusual or unexpected
- Outliers can be dangerous (later we will explain why)



Scatter Plot: Outliers

Example: Austin housing data set



Scatter Plot: Outliers

Example: Austin housing data set (This is how outliers look!)

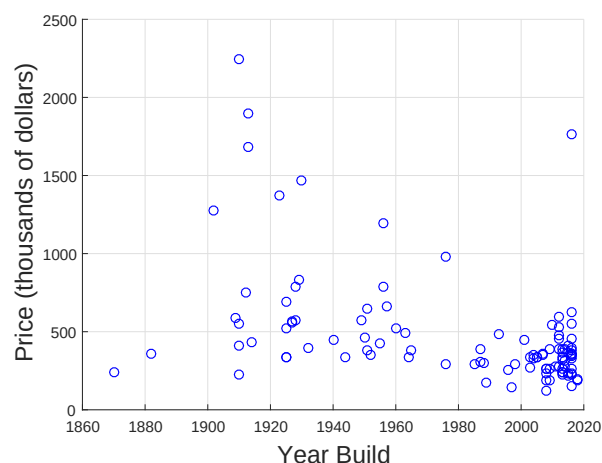


Scatter Plot: Skewness

Skewness: When one of the variables (or both) are skewed, then it might be difficult to analyze visually

Example: You want to see relationship of house prices and year of construction in Austin

- Year of construction (skewed to the left): Most houses are new
- House prices (skewed to the right): Most houses cost less than \$500,000

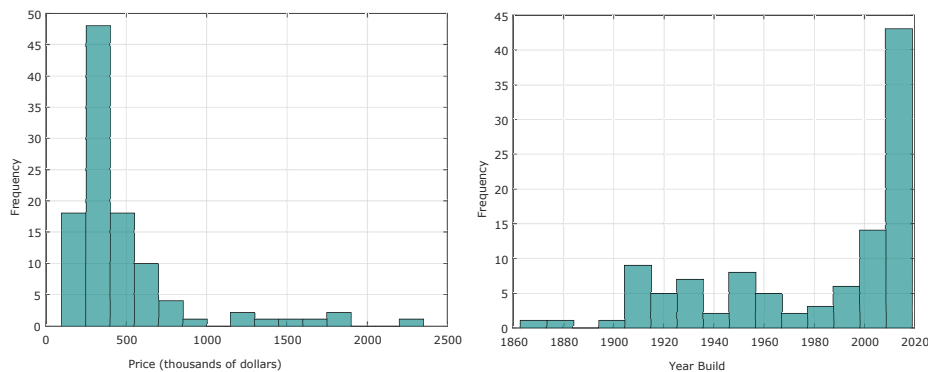


Scatter Plot: Skewness

Skewness: When one of the variables (or both) are skewed, then it might be difficult to analyze visually

Example: You want to see relation between house prices and year of construction in Austin

- Year of construction (skewed to the left): Most houses are new
- House prices (skewed to the right): Most houses cost less than \$500000

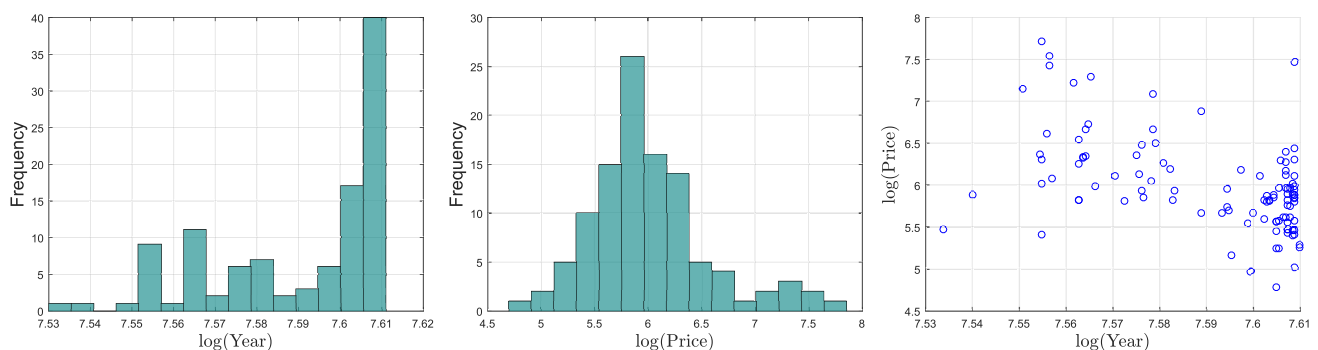


Transformation: Skewness (Not Tested)

Skewness can be partly remedied by performing a log transformation on the data

Example: Instead of plotting (Year) v.s. (Price),

- Plot $\log(\text{Year})$ v.s. $\log(\text{Price})$



The log transformation allows us to see somewhat better the relationship

Coefficient of Correlation

Coefficient of Correlation:

- is denoted by r
- measures the strength of the *linear association* between two variables x and y

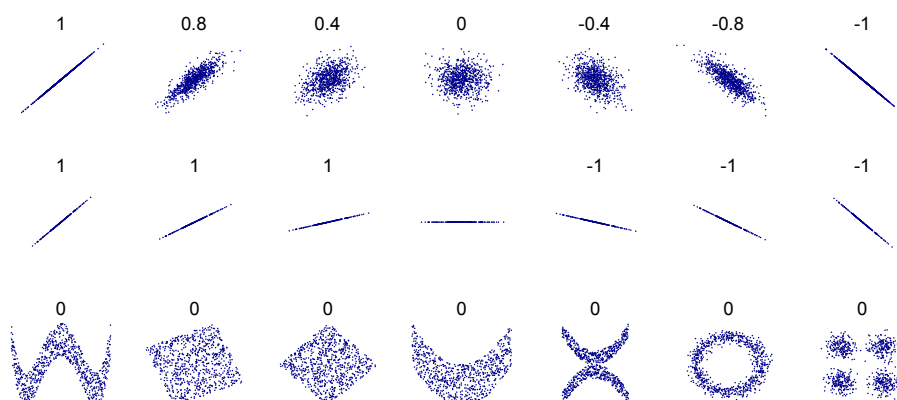
Properties:

- r is always between -1 and $+1$
- If $r > 0$, then correlation is positive
- If $r < 0$, then correlation is negative
- The closer to $+1$, the stronger the positive relationship between x and y
- The closer to -1 , the stronger the negative relationship between x and y
- r does not depend on the units of measurement
- r is only useful for linear relationships

Coefficient of Correlation

Properties:

- The closer to $+1$, the stronger the positive relationship between x and y
- The closer to -1 , the stronger the negative relationship between x and y

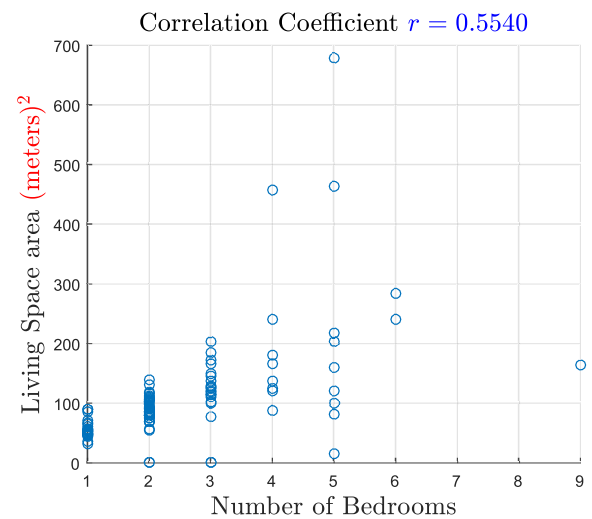
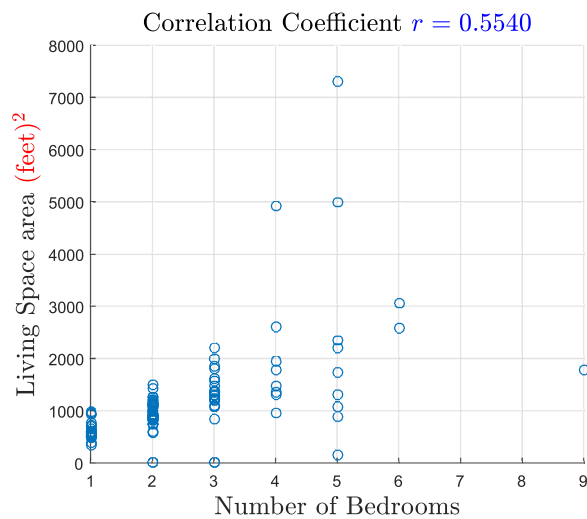


- r is only useful for linear relationships

Coefficient of Correlation

Properties:

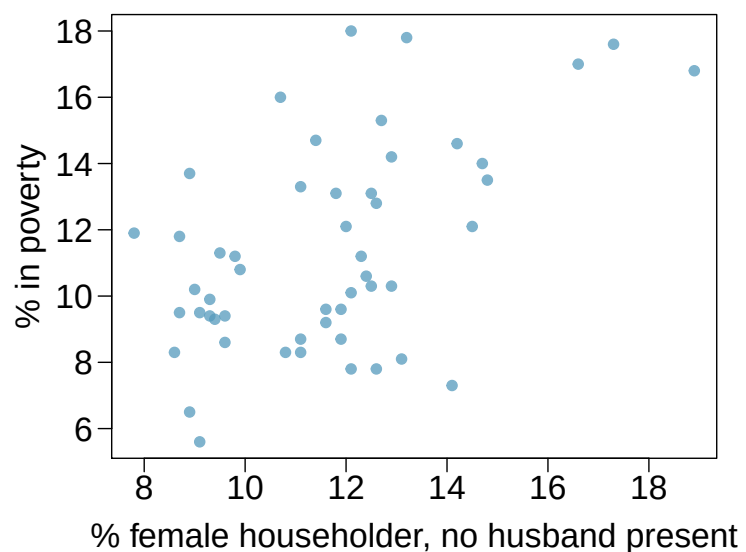
- r does not depend on the units of measurement



Practice: Scatterplot & correlation

Which of the following is the best guess for the correlation between % in poverty and % no husband in the household?

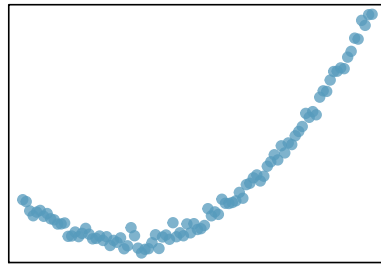
- A 0.1
- B -0.6
- C -0.4
- D 0.9
- E 0.5



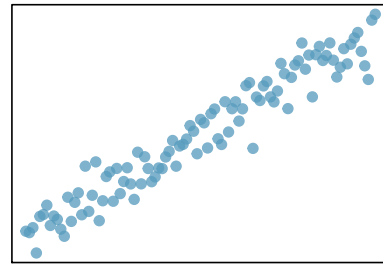
Answer: E.

Practice: Visualize the correlation

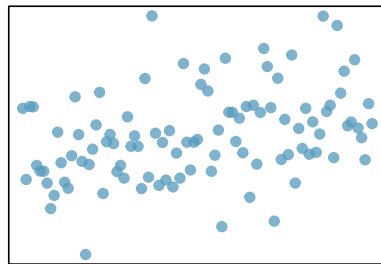
Which of the following has the strongest correlation, i.e. correlation coefficient closest to $+1$ or -1 ?



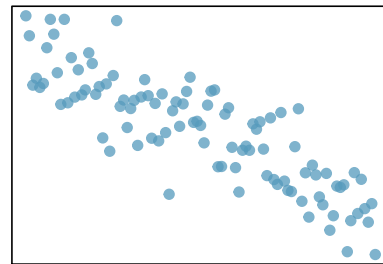
(a)



(b)



(c)



(d)

Answer: (b) → correlation means linear association.

Takeaway: ALWAYS Visualize your Data FIRST



<https://www.theguardian.com/lifeandstyle/2008/nov/23/active-games-kids-guide-blindman-s-buff>

Before Next Lecture

- Be able to import data from excel to python.
- Calculate basic statistics for *Darts.csv* file
- *Syllabus Quiz* due on 11:59PM Sep. 6
- Sign up teams for project

